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ORIGINAL RESEARCH

A Case-Driven and Knowledge Graph-Supported Approach to Pathology Education: Effects on Knowledge Retention and Clinical Reasoning

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Objective: Given the issues of knowledge fragmentation and insufficient clinical reasoning associated with traditional linear pathology teaching, this study developed and evaluated a "case-driven, knowledge-graph-supported" curriculum restructuring model. The model was designed to shift the instructional approach from a discipline-centric logic toward a clinical diagnostic reasoning framework.

Methods: Fifty-four five-year clinical medicine undergraduates at Peking University Health Science Center participated. Using clinical cases as the organizing framework, the "Urinary System Pathology" course was restructured into a knowledge graph-supported progressive learning framework. A mixed-methods evaluation design was adopted to comprehensively assess teaching effectiveness, including three in-session formative tests, a 14-day delayed knowledge retention assessment, and a structured student learning experience questionnaire.

Results: Students showed significant improvements across the three immediate in-session formative tests, with median percentage scores of 76.6%, 87.8%, and 85.9%, respectively ($P < 0.001$). In the day-14 retention test, the median percentage score under open-book conditions was higher than the average of the three in-session tests (91.8% vs 83.5%, $P < 0.001$). Additionally, the closed-book median percentage score was also higher than that of the three in-session tests (89.0% vs 83.5%, $P = 0.016$), with no significant difference found between open- and closed-book conditions ($P = 0.69$). Students' self-rated competency in renal disease diagnosis showed a marked increase, with the average score rising from 2.0 to 4.4 ($P < 0.001$). However, alongside these positive learning outcomes, the course also presented notable challenges for a subset of learners. 14.8% of participants reported challenges related to course pace and excessive cognitive load during the learning process.

Conclusion: The proposed model was associated with improved student performance in our cohort. These preliminary findings suggest potential benefits for addressing knowledge fragmentation and inert knowledge in pathology education. However, controlled studies are needed to establish causality. In addition, the pacing challenges identified in this study highlight the need for continued curriculum refinement to balance the benefits of case-based learning against the cognitive demands it places on students. Future studies should also adopt multi-center longitudinal designs with control groups to further examine the long-term learning transfer and sustained educational effects of this teaching model.

Keywords: pathology education, curriculum reconstruction, case-driven learning, knowledge graph, clinical reasoning

Introduction

As a core bridge between basic and clinical medicine, pathology aims to build a coherent causal pathway linking disease mechanisms to clinical manifestations.[1] Traditional pathology teaching, organized around disciplinary divisions, has long adopted a linear structure (eg, presenting etiology, pathogenesis and pathological changes in a fixed sequence). Cognitively, this traditional framework has several limitations. First, course content is structured based on discipline--based classification rather than clinical problem-solving logic, which tends to separate concepts that are closely related in

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clinical practice. Second, one-way lecture-based teaching provides few opportunities for students to actively develop their own cognitive frameworks for clinical reasoning. Together, these drawbacks contribute to two key problems: fragmented knowledge and inert knowledge. Consequently, students often struggle to apply theoretical knowledge to real clinical scenarios.[2]

To address these limitations, this study introduces a case-centered course framework for pathology education, grounded in three complementary learning theories. First, constructivism provides the foundation for understanding how students learn; it emphasizes that knowledge is actively built through problem-solving, rather than passively memorized as isolated facts.[3] Second, situated learning theory suggests that separating pathology from clinical cases contributes to inert knowledge.[4] Although Case-Based Learning (CBL) was introduced to bridge this gap, conventional CBL often tends to become a superficial process in which cases are treated as post-class supplements to textbook facts.[5,6] Furthermore, the temporal and spatial separation between lectures and discussions may create a disconnect in learning, requiring students to struggle with memory decay. Therefore, to shift the learning logic from textbook chapters to clinical reasoning requires elevating clinical cases to the organizing thread of knowledge. Third, because navigating these real cases requires intense thinking, cognitive load theory (CLT) provides the principles for managing this mental workload and minimizing student overload.[7,8] Under traditional linear teaching, scattered facts have high element interactivity, creating a heavy intrinsic cognitive burden without helping students build structured mental frameworks.[9] To address this, knowledge graph (KG) technology can be introduced as a powerful visual support tool that explicitly organizes knowledge through semantic network modeling, thereby reducing cognitive overload and integrating fragmented content. First, by explicitly constructing "entity-relation" semantic networks, KG transforms fragmented, isolated knowledge points into an interconnected, holistic structure. This counters the problem of knowledge fragmentation by visually demonstrating the causal, hierarchical, and sequential relationships between concepts (eg, linking a specific pathological change to its etiology and clinical manifestation). Second, from the perspective of cognitive load theory, the high element interactivity inherent in pathological reasoning is a major source of intrinsic cognitive load. A KG mitigates this by providing a visual schema that reorganizes these complex, high-interactivity elements into a single, low-interactivity visual representation.[10] This scaffolding assists students in instantly integrating fragmented case information into interconnected knowledge networks.[11]

To operationalize and validate these theoretical frameworks, this study developed a systematic, case-driven curriculum restructuring model supported by a KG. The primary goal was to break away from conventional, textbook-aligned linear instruction and reorient the curriculum around authentic clinical diagnostic reasoning. By embedding this model into a standard undergraduate medical curriculum, the study may serve as a reference for educators seeking to address the persistent challenge of inert knowledge and promoting higher-order clinical reasoning.

Materials and Methods

Construction Pathway of the Multimodal KG

This study constructed a multimodal KG for the pathology curriculum on the Wisdom platform, a Chinese online education platform that provides built-in tools for KG construction and multimodal resource linking. The construction process was operationalized as follows: (1) Construction of the foundational declarative KG (knowledge-oriented). First, based strictly on the pathology syllabus, the research team extracted and established core knowledge elements to serve as the skeletal nodes of the graph. The curriculum content was systematically decomposed into 105 knowledge points. Through semantic annotation and hierarchical organization, a total of 1490 knowledge nodes were identified. For instance, nodes such as "acute inflammatory response", "granulation tissue", and "crescentic glomerulonephritis" were defined. This was performed manually using the platform's node creation interface, with each node tagged by its corresponding chapter and topic. Subsequently, multimodal instructional resources, including 68 micro-lecture videos, 91 PPTs, 201 high-resolution digital slides, and 690 exercises, were manually linked via the platform's resource management interface to their corresponding knowledge nodes. Finally, using the platform's built-in semantic linking function, a foundational knowledge network was established, wherein knowledge nodes functioned as entities and logical semantics, such as pathological causality ("cause-of") and clinical manifestation ("manifests-as"), constituted the edges.

(2) Construction of the intermediate KG (problem-oriented). This layer addresses the linear structure of traditional textbooks by reorganizing pathological mechanisms and morphological features around core clinical manifestations, using these manifestations as problem anchors. A total of 41 clinical problems were formulated as the core anchors of this layer. This design enables cross-linking both within and across chapters. For example, the question "How to distinguish chronic glomerulonephritis from chronic pyelonephritis?" is linked to the relevant nodes from the foundational declarative KG. These problem-oriented links guide students to retrieve and integrate fragmented declarative knowledge from the foundational layer in order to address specific clinical puzzles, thereby facilitating the transition from rote memorization to analytical reasoning. (3) Construction of the advanced KG for higher-order capability. This level was built upon a set of 29 real clinical cases or clinical application scenarios (eg, "A 48-year-old male patient developed bilateral ankle pitting edema and markedly increased foamy urine approximately 7 months ago"). Students were then explicitly guided to analyze the possible pathological changes, differential diagnosis and their underlying mechanisms based on the clinical presentation. This case-driven analysis was not conducted in isolation. Instead, it was linked to the corresponding problems in the intermediate KG, and through that, access the specific knowledge nodes in the foundational declarative KG. This three-tiered design (Knowledge-based network → problem-oriented →Applied-lev el, Supplementary Figure 1) allows students to embed the pathological mechanisms of a specific case into a broader conceptual framework.

Instructional Design and Implementation Pathway

Curriculum Restructuring Logic

This study replaces the traditional, linear, discipline-centered logic of pathology, which sequentially presents "concept, etiology, pathogenesis, pathological change, and clinical manifestation", with a case-driven, KG-supported teaching model. Utilizing authentic clinical cases as the logical entry point, this model guides students to actively extract and construct interconnected knowledge networks while engaging in clinical diagnostic-solving. To rigorously test the model under conditions of high cognitive load, the curriculum intervention was implemented within the "Urinary System Pathology" module. This module, particularly the section on glomerulonephritis, inherently imposes the highest intrinsic cognitive load across the pathology curriculum due to its highly abstract pathogenetic mechanisms and sophisticated morphological diagnostic criteria.

Participants and Instructors

The participants comprised 54 third-year medical students from an intact class within the 2024 five-year clinical medicine program at Peking University Health Science Center. To control for implementation bias, all instructional sessions were delivered by the same senior pathology instructor, who possesses over 10 years of experience in both pathology education and clinical pathological diagnosis of urinary system diseases. Written informed consent was obtained from all individual participants included in the study.

Implementation of the Restructured Curriculum

This instructional unit comprised a total of 6 class hours, delivered across three progressive sessions (2 class hours per session, 50 minutes per class hour). Each session followed a standardized three-stage process: pre-session autonomous preparation, in-session case-driven instruction, and post-session reinforcement and extension. Figure 1 depicts the overall framework of the reconstructed pathology curriculum. To demonstrate how this model covers the traditional syllabus content, the specific knowledge points and their restructured instructional logic are detailed in Supplementary Table 1.

Session 1: Glomerulonephritis (2 Class Hours)

① Pre-session Autonomous Preparation: The instructor distributed pre-session tasks via the KG platform. Students conducted self-directed learning supported by the foundational KG, where core knowledge nodes were linked to multimodal resources such as micro-lectures and digital slides to facilitate the construction of a preliminary knowledge framework.

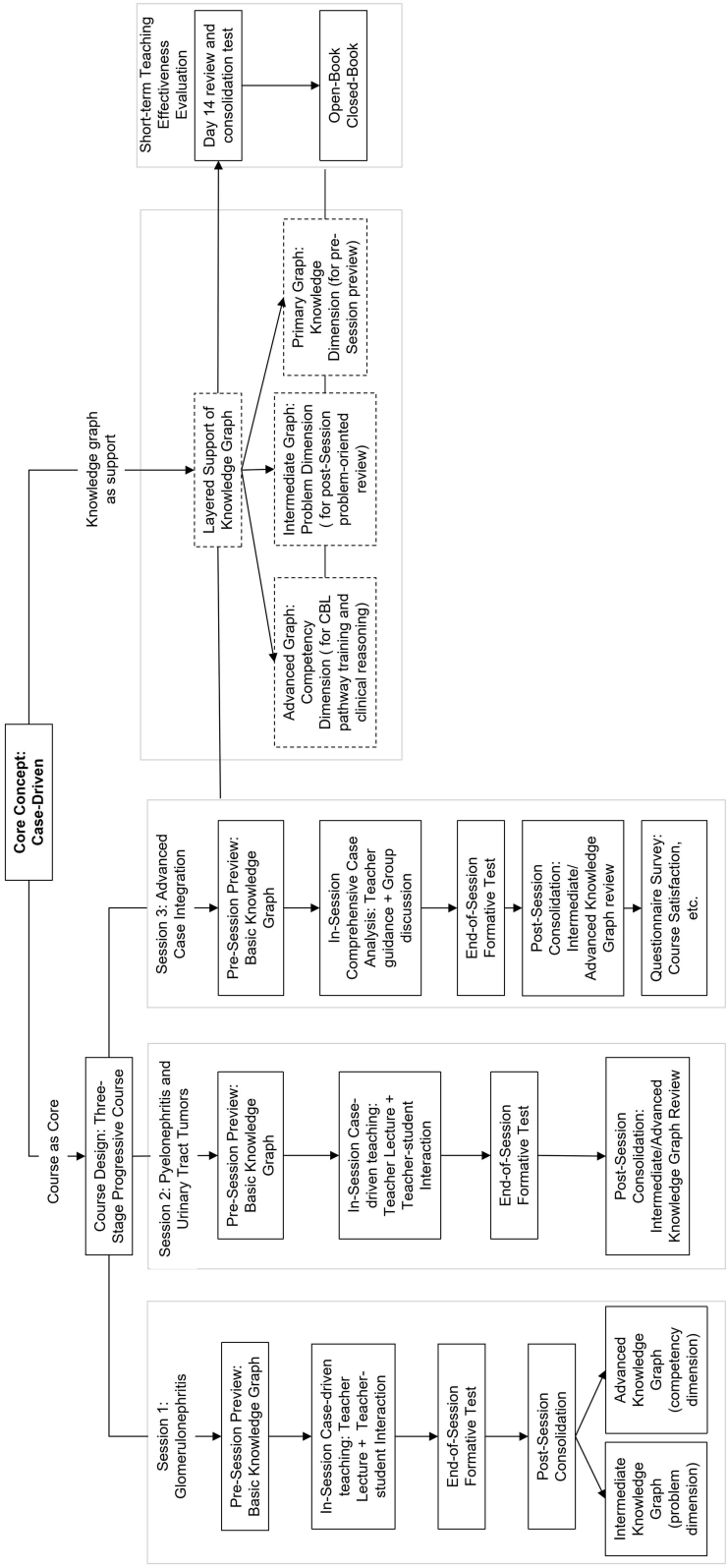


Figure 1 Overall framework of the reconstructed pathology curriculum.

② In-Session Case-Driven Instruction: Case-driven instructional design was executed using four representative clinical cases: minimal change disease (MCD), membranous nephropathy (MN), acute endocapillary proliferative glomerulonephritis, and crescentic glomerulonephritis. Taking MCD as an example, the instructor presented a clinical case and raised six progressive questions that systematically guided students from clinical diagnosis to pathological changes, clinicopathological correlation, etiology and pathogenesis, treatment principles, and prognosis, thereby embedding the syllabus knowledge points within a clinical reasoning sequence (details are provided in the Supplementary Methods). After completing each case, the instructor utilized the KG to synthesize and cross-link core concepts, helping students construct a structured internal knowledge network. The remaining three cases followed a similar procedure, where students were further guided to comparatively analyze the similarities and differences among various pathological phenotypes.

③ Post-Session Reinforcement and Extension: Students accessed the intermediate (problem-oriented) and advanced (capability-oriented) KGs for review. The intermediate graph featured layered queries centered on core clinical issues such as "diagnosis and differential diagnosis of nephrotic syndrome", while the advanced graph cultivated students' clinical reasoning skills through complete, longitudinal CBL case pathways.

Session 2: Pyelonephritis and Renal Tumors (2 Class Hours)

The structure followed that of Session 1, utilizing the same three-stage process. Four cases –pyelonephritis, clear cell renal cell carcinoma (ccRCC), nephroblastoma (Wilms' tumor), and urothelial carcinoma, were employed for case-driven instruction. Pre-session and post-session activities were identical to those described in Session 1.

Session 3: Advanced Cases and Comprehensive Case Analysis (2 Class Hours)

The three-stage structure was again followed. Utilizing three advanced cases-membranoproliferative glomerulonephritis (MPGN), IgA nephropathy, and chronic kidney disease (CKD) complicated by renal cell carcinoma, this session combined instructor guidance with small-group discussions to perform comprehensive case analysis and differential diagnosis. Pre-session and post-session activities were identical to those described in Session 1.

Evaluation Methods and Implementation Procedures

To comprehensively and objectively evaluate the educational efficacy of this restructured model on students' knowledge retention, this study established a mixed-methods evaluation framework spanning two distinct time points: immediate and delayed assessments. This approach combined standardized testing (Supplementary Table 2) with a student experience questionnaire. The specific components are as follows: (1) In-session formative testing (Immediate assessment): Executed at the end of each session. Each quiz comprised 5 text-only multiple-choice questions (MCQs, specifically A1/A2 types, 2 points per question). (2) Delayed retention testing: Administered 14 days after the conclusion of the entire instructional unit, consisting of two components: ① Open-book MCQ test: Included 25 A1/A2-type MCQs (2 points per question). During the test, students were permitted to consult the multimodal KG to assess their resource-assisted information retention and retrieval proficiency. ② Closed-book clinicopathological analysis test: Conducted following the open-book test, comprising 2 subjective case-analysis questions (30 points per question). This assessment strictly enforced a closed-book protocol, designed to evaluate whether students could autonomously integrate information and execute clinical reasoning after the complete fading of external cognitive scaffolds. All test items (both in-session testing and delayed retention testing) were drawn from a centralized item bank developed by the same teaching team based on the unified course learning objectives, which covers all core knowledge points and is categorized by cognitive level according to Bloom's taxonomy.[12] The items in this bank have been validated through previous final examinations, with difficulty indices ranging from 0.3 to 0.8 and discrimination indices exceeding 0.25. Each test was a new set of questions. The in-session testing targeted the core content of each respective session, while the delayed open-book MCQ testing comprehensively covered the core content from all three sessions. To ensure comparability across assessments, the proportion of easy, medium, and difficult items was held constant across all tests. Difficulty was classified by the item difficulty index (P) as follows: easy items ($0.65 < P \leq 0.80$), medium items ($0.45 < P \leq 0.65$), and difficult items ($0.30 < P \leq 0.45$). All tests maintained the same ratio of items across these three difficulty levels. In addition, the two subjective

case-analysis questions were both set at a medium difficulty level. (3) Questionnaire survey: Distributed upon the completion of the third session. The questionnaire utilized a 5-point Likert scale for anonymous rating, with its reliability verified by Cronbach's α coefficient. The items captured subjective student feedback across dimensions including the stimulation of learning interest, enhancement of clinical reasoning capabilities, and overall satisfaction with the model. Crucially, retrospective pre-test items (eg, "Prior to this course, I rated my ability to diagnose common urinary system diseases as: 1–5 points") were embedded within the instrument to capture evidence of students' self-perceived knowledge gains. Students filled it out voluntarily and anonymously, with settings allowing only one submission per IP address. Questionnaires not fully completed were considered invalid.

Grading Process and Reliability Control for Subjective Questions

To guarantee the objectivity and stability of grading for the subjective clinicopathological analysis questions, this study utilized a structured rubric and a standardized grading workflow. First, questions were scored using a structured rubric including three scoring units: pathological diagnosis (20 points) assessing the accuracy of the diagnostic term and the supporting rationale, pathogenesis (20 points) evaluating the identification of core mechanisms and the integration of the reasoning chain from etiology to pathology, and clinicopathological correlations (20 points) measuring the ability to link pathological findings to clinical symptoms and treatment principles, as well as the accuracy of the applied knowledge. Each dimension was further subdivided into specific scoring units with detailed criteria for different performance levels (Supplementary Table 3). Second, two raters with both pathology teaching and extensive clinical pathological diagnostic experience collaboratively reviewed the rubric and independently graded 3 to 5 randomly selected pilot answer sheets to calibrate the grading standards. Subsequently, both raters conducted independent, blind grading of all student answer sheets. Finally, the Intraclass Correlation Coefficient (ICC) was employed to calculate the inter-rater reliability based on the total scores awarded by the two raters, with an ICC value ≥ 0.75 established as the threshold for good reliability. If the ICC fell below this threshold, the two raters discussed the points of divergence to reach a consensus through negotiation. The mean score from both raters was adopted as the final grade.

Statistical Analysis

Statistical analysis was performed using SPSS version 26.0. To standardize the scoring scales across different tests and facilitate meaningful comparisons, all raw test scores were converted into percentage scores. Specifically, raw scores were standardized as a percentage of the total points within each level. Given that the percentage scores did not meet the assumption of normality across different assessment time points, non-parametric tests were uniformly adopted to enhance the robustness of the statistical outcomes. The overall comparisons of percentage scores across multiple time points were conducted using the Friedman test. If the Friedman test yielded significant results ($P < 0.05$), post hoc pairwise comparisons were further executed utilizing the Wilcoxon signed-rank test. To control the inflated risk of Type I errors arising from multiple comparisons, the significance levels were adjusted using the Bonferroni correction. For the questionnaire analysis, a paired *t*-test was employed to analyze changes in students' self-perceived capabilities.

To provide a more nuanced understanding of the depth of learning, we classified the test items into two cognitive levels based on the revised Bloom's taxonomy: "remembering/understanding" (basic recall and comprehension of pathological concepts) and "applying/analyzing" (higher-order cognitive tasks requiring application of knowledge to clinical scenarios and analysis of pathophysiological mechanisms). This classification was performed by two independent researchers and any discrepancies were resolved through discussion.

Results

Students' Learning Outcomes and Knowledge Retention

Immediate Knowledge Acquisition: Longitudinal Comparison of Three In-Session Formative Tests

To evaluate students' knowledge acquisition and immediate application capabilities during the course, this study analyzed the median percentage scores of the three in-session formative tests, which were 76.6%, 87.8%, and 85.9%, respectively. A statistically significant difference was observed among the three median scores (Friedman $\chi^2 = 18.8$, $P < 0.001$). Post hoc

Table 1 Performance Comparison Between In-Session Formative Tests and the Day 14 Delayed Test

Evaluation Parameter	In-Session Formative Tests (Three Sessions Combined)	Consolidation Test (Open-Book)	Consolidation Test (Closed-Book)	*P*
Percentage scores [median (range)]	83.5% (53.3–100.0%)	91.8% (76.0–100.0%)	89.0% (66.7–100.0%)	<0.001
Remembering & Understanding [median (range)]	86.7% (50.0–100.0%)	95.4% (83.3–100.0%)	–	<0.001
Analyzing & Applying [median(range)]	81.7% (44.4–100.0%)	88.5% (69.2–100.0%)	89.0% (66.7–100.0%)	0.006

pairwise comparisons indicated that the median percentage score of the second in-session test was significantly higher than that of the first test (Z = −3.3, *adjusted P* = 0.003). However, after Bonferroni correction, the difference in median scores between the third and first tests reached only borderline statistical significance (Z = −2.4, *adjusted P* = 0.05). No statistically significant difference was detected between the median scores of the third and second in-session tests (Z = 1.0, *adjusted P* = 0.9).

Day 14 Retention Test: Short-Term Knowledge Retention and Sustained Application Ability

We conducted two follow-up assessments 14 days after course completion, an open-book MCQs test and a closed-book clinicopathological analysis test, to evaluate short-term knowledge retention and application performance with or without reference materials. Outcomes were compared with the aggregated performance of the three in-session tests. The results demonstrated a statistically significant difference in students' median percentage scores across the three assessment conditions (Table 1; Friedman $\chi^2 = 18.0$, $P < 0.001$).

Post hoc pairwise comparisons revealed that the median percentage score under the Day 14 open-book condition (91.8%) was significantly higher than that of the in-session immediate tests (83.5%, *adjusted P* < 0.001). Notably, the median score under the Day 14 closed-book condition (89.0%) was also significantly higher than the in-session performance (83.5%, *adjusted P* = 0.016). No statistically significant difference in median scores was detected between the Day 14 open-book and closed-book conditions (*adjusted P* = 0.69).

To further elucidate educational efficacy at distinct cognitive levels, test items were categorized by cognitive level ("remembering/understanding" and "applying/analyzing"), followed by a subgroup analysis of the median percentage scores for each level across the assessment time points. The analysis indicated that at the remembering/understanding level, the median score under the Day 14 open-book condition (95.4%) was significantly higher than that of the in-session immediate tests (86.7%, *adjusted P* < 0.001).

For higher-order cognitive tasks (applying and analyzing), a statistically significant difference was also observed in the median percentage scores across the three conditions (Friedman $\chi^2 = 10.1$, $P = 0.006$). Post hoc pairwise comparisons demonstrated that the median score under the Day 14 open-book condition (88.5%) was significantly higher than that of the in-session immediate tests (81.7%, *adjusted P* = 0.018). Mirroring this trend, the median score under the Day 14 closed-book condition (89.0%) was also significantly superior to the in-session performance (81.7%, *adjusted P* = 0.028). Crucially, no statistically significant difference in median percentage scores was detected between the Day 14 open-book and closed-book conditions at this high-order cognitive level (*adjusted P* = 1.0).

Student Perceptions of the Learning Experience

Retrospective Self-Assessment

The paired samples *t*-test revealed that in the retrospective self-assessment of "the ability to diagnose common urinary system diseases", students' self-rated scores significantly increased from a pre-course mean of 2.0 (SD = 1.1) to a post-course mean of 4.4 (SD = 0.7) with a statistically significant difference (t = −15.4, $P < 0.001$).

Learning Experience Questionnaire

A total of 47 valid questionnaires were recovered upon completion of the course (response rate: 87.0%). The Cronbach's α coefficient was 0.83, suggesting good internal consistency. Students' ratings across all dimensions are presented in Table 2. Overall, students expressed high satisfaction with the new teaching model, with 78.7% selecting "highly satisfied". A large majority (85.1%) strongly agreed that this model outperformed traditional lectures, and 70.2% reported

Table 2 Student Feedback on the Case-Driven and KG-Supported Pathology Model (n = 47)

Survey Item	Strongly Agree [n(%)]	Agree [n(%)]	Neutral [n(%)]	Disagree [n(%)]	Strongly Disagree [n(%)]
I am satisfied with the overall instructional effectiveness of this course.	37 (78.7%)	10 (21.3%)	0 (0)	0 (0)	0 (0)
This course stimulated my interest in learning pathology.	33 (70.2%)	14 (29.8%)	0 (0)	0 (0)	0 (0)
The workload and instructional pace of this course were moderate and acceptable	19 (40.4%)	21 (44.7%)	6 (12.7%)	1 (2.1%)	0 (0)
This course effectively enhanced my capability to analyze cases and perform clinical reasoning.	33 (70.2%)	14 (29.8%)	0 (0)	0 (0)	0 (0)
This course inspired my awareness to autonomously investigate unresolved problems.	22 (46.8%)	20 (42.6%)	5 (10.6)	0 (0)	0 (0)
The KG was highly beneficial in helping me map out the intrinsic logic and correlations of disease knowledge.	22 (46.8%)	18 (38.3%)	7 (14.9%)	0 (0)	0 (0)
This instructional model is more effective than the traditional lecture-based model.	40 (85.1%)	7 (14.9%)	0 (0)	0 (0)	0 (0)

that it greatly boosted their interest in pathology learning. Likewise, 70.2% strongly agreed that the approach improved their skills in case analysis and clinical reasoning. In addition, 89.4% of respondents stated that the course enhanced their initiative to explore unresolved questions independently.

As an instructional scaffold, the KG functioned as intended. 85.1% of students agreed that it helped them consolidate the logical relationships and connections between disease-related knowledge.

Nevertheless, opinions varied regarding course workload and teaching pace. While 83.0% of participants generally accepted the current arrangement, only 40.4% chose "strongly agree". 12.7% (6 students) selected "neutral", and 2.1% (1 student) indicated "disagree".

Discussion

This study implemented and evaluated a case-driven, KG-supported curriculum restructuring model, utilizing urinary system pathology as the contextual framework. In this model, knowledge points were rooted within the logic of clinical diagnosis and treatment. We observed that, as the course progressed, students not only achieved improvements in immediate knowledge acquisition but also demonstrated enhanced short-term knowledge retention at Day 14 under both open-book and closed-book conditions. This teaching model was well received by the students, which may offer a reproducible framework for case-driven and technology-enhanced medical education.

Effects of the Case-Driven and KG-Supported Model in Knowledge Acquisition

Prior work has demonstrated the feasibility of integrating KGs into medical education. Based on our literature review, most existing KG constructions have remained primarily at the declarative knowledge level, focusing on structuring course content and visualizing relationships among knowledge points.[11,13,14] A small number of studies have attempted to build problem and competency layers in addition to the declarative knowledge level.[10,15] For instance, Wei et al[10] have explicitly constructed a hierarchical "knowledge-problem-competency" KG that extends beyond knowledge organization to model problem-oriented reasoning and clinical competency development in pathology teaching. Our KG construction adopts a similar three-tiered architecture. This layer design is intended to enable students to progressively move from memorizing fragmented declarative knowledge, to analyzing mechanism-based clinical problems, and ultimately to engaging in integrated clinical reasoning within a coherent conceptual framework.

KGs have been used as supplementary tools across pre-class, in-class, and post-class phases,[10,11,13,14] typically integrated within existing chapter-based curricula to support preview, instruction, review, and resource recommendation. Although our study shares a similar technical basis and workflow, it differs in how the curriculum is organized.

According to situated learning theory, separating pathological mechanisms from clinical realities tends to produce inert knowledge; thus, instructions grounded in real patient scenarios are essential for cultivating clinical reasoning.[2,16,17] However, the manner in which clinical cases are integrated into the curriculum varies substantially across studies, and these differences have important implications for the depth and nature of student learning. In prior studies in pathology teaching, the implementation of case-based instruction has predominantly followed two patterns. First, cases are used as supplementary materials or testing tools after theoretical lectures.[14] Second, cases serve as the materials of in-class discussion or activities,[18] but the overall curriculum structure remains primarily organized around textbook chapters. Cases are often embedded within this chapter-based framework as tools for stimulating interest, discussion, or reasoning, rather than as the fundamental organizing principle of the curriculum itself. In contrast, our study reorganizes the curriculum around clinical cases as the primary organizing framework, rather than being driven primarily by textbook chapters. The learning sequence begins with a clinical case presentation, guides students through reverse reasoning from symptoms to pathological changes and mechanisms. During this process, students encounter and learn specific knowledge points as they are needed to solve the clinical problem at hand. This represents a move toward a curriculum where cases play a more central organizing role in pathology teaching.

The teaching model achieved favorable educational outcomes, as reflected in student performance across all assessments. The statistically significant improvement from the first to the second formative test suggests that students rapidly adapted to the course format. Although the difference between the first and third tests showed only borderline significance, this finding is likely attributable to the curricular design of the third session, which comprised advanced cases and comprehensive clinical scenarios requiring higher-order synthesis. In other words, the increased cognitive load imposed by the third session may have offset the expected upward trend in scores. The delayed retention test on Day 14 post-course further provides crucial empirical evidence. Under the closed-book condition where the KG was completely removed, students' median percentage score at the analyzing and applying level (89.0%) demonstrated no statistically significant difference compared to the open-book condition with KG support (88.5%), and both were significantly superior to the immediate in-session performance (81.7%). These results suggest that the observed learning gains were sustained over time, rather than transient memorization. Moreover, students' performance may indicate a progressive shift from resource-dependent performance to more autonomous knowledge retrieval.

Higher-Order Capacity Gains and Its Accompanying Pacing Challenges

The questionnaire findings reveal a discrepancy between students' perceived learning gains and their classroom experience. On one hand, students widely agreed that the curriculum stimulated their interest and improved their clinical reasoning skills. However, this positive feedback contrasted with their evaluation of the instructional pacing: only 40.4% strongly agreed that the workload and pace were moderate, while 14.8% felt neutral or negative. To examine the sources of these pacing difficulties, we consider two interrelated factors: (1) in-session cognitive demands, and (2) the density of in-session activities. By anchoring learning within clinical contexts, this model organizes complex pathology knowledge into a clinical framework, saving students from struggling with disconnected textbook concepts. However, this design simultaneously introduces a new kind of cognitive demand. Because classroom activities focus on higher-order clinical reasoning, students must constantly perform intense mental tasks, such as extracting key data, reorganizing knowledge, and making clinical analysis. This sudden shift from passive listening to continuous high-level thinking increases task complexity. The density of in-session activities further compounds these demands. Each session is structured around multiple case-based problems that require sustained active participation. The cumulative effect above may be the root cause of why a subset of students felt the class pace was too fast and the workload was overwhelming.

This phenomenon is by no means unique to our model; it is a common challenge in educational reforms that target high-order thinking. Studies on TBL,[19] and PBL/CBL[20] have consistently reported that while these active learning methods successfully improve higher-order skills, they also impose increased cognitive load on students due to the intensive collaboration, reasoning, and self-directed information synthesis they require. The pacing issues observed in our model reflect this same pattern within the specific context of pathology education. However, it is noteworthy that students' concerns about the fast pace did not reduce their overall satisfaction with the course. In fact, all surveyed students still felt that this model was better than traditional lecturing, and unanimously agreed that it stimulated their

interest and improved their clinical reasoning skills. This shows that despite the heavy mental effort required in each session, students could clearly see the value of their hard work. In educational psychology, evidence shows that when learners view difficult tasks as necessary for real progress, the high perceived value of the learning (such as interest and usefulness) can easily outweigh the costs, such as time pressure and hard effort.[21] Therefore, this balance between high difficulty and high satisfaction provides a valuable foundation for improving our teaching strategies in the future.

Instructional Optimization: Managing Cognitive Demands Across Teaching Phases

Based on CLT and our survey findings, future teaching improvements should focus on managing students' mental workload through the following steps. To address the density of in-session activities and the sustained high-level cognitive engagement, we can adopt a step-by-step approach that combines short lecture segments with immediate practice. The teaching process for each clinical case can be divided into several small parts, each lasting 10 to 15 minutes. After completing each part, such as discussing a clinical diagnosis or a pathological mechanism, a short quiz or activity (3 to 5 minutes) will be introduced. This could involve using one or two MCQs to check understanding, or asking students to explain the key ideas to their peers. This alternating lecture-and-practice design serves two main purposes: (1) it breaks down complex cases into smaller, manageable pieces, lowering the information density and making the session feel less rushed; (2) these quick exercises leverage the testing effect to consolidate learning and reduce cognitive overload.[22]

Second, to address the varying levels of student preparedness and ability, which may influence how individual students experience pre-session preparation and in-session cognitive demands, we can use KGs to provide personalized learning paths and different levels of support based on each student's current ability. For high-performing students, the system will provide access to challenging cases from the advanced KG. In contrast, for students who struggle, it will prioritize simpler cases. This setup ensures that the task difficulty matches each student's actual level, preventing cognitive overload.[23]

Limitations and Future Directions

There were several limitations in this study. First, this study adopted a single-group, longitudinal design and lacked a fully parallel, blinded control group undergoing traditional instruction, which inherently weakens the strength of causal inference regarding the model's independent effect. Second, the study sample was derived from a single natural class (n = 54) at a single medical institution. The relatively small sample size necessitates further multi-institutional validation to confirm the generalizability of this model. Finally, the delayed retention test on Day 14 post-course only captures short-term knowledge retention. Whether this model exerts a sustained positive impact on medical students' long-term transfer of learning, specifically, their long-term clinical reasoning performance during subsequent clinical clerkships and hospital rotations, remains to be verified by longitudinal studies with extended follow-up periods.

Conclusion

The long-standing dilemma in pathology education stems largely from a mismatch between how knowledge is organized by discipline and how students naturally process information in clinical contexts. This study examined the value of a "case-driven and KG-supported" teaching model that changes clinical cases from supplementary exercises into the core framework that guides the entire learning process, thereby shifting the teaching focus from "discipline-centric" to "clinical-reasoning" logic.

We observed that this new approach was associated with improved high-order cognitive performance among medical students in our cohort. Notably, students were able to independently apply their clinical reasoning skills even after the KGs and instructor guidance were completely removed. However, a controlled study is required to determine whether the intervention itself caused the observed improvement. In addition, further longitudinal research is needed to manage cognitive demands across different learning phases and to assess long-term retention.

Data Sharing Statement

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Human Ethics and Consent to Participate Declarations

This study was approved by the Biomedical Ethics Committee of Peking University (Approval Number: IRB00001052-25169), and the research procedures strictly adhered to the ethical principles for medical research established by the Declaration of Helsinki.

Consent to Participate

Written informed consent was obtained from all individual participants included in the study.

Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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Disclosure

All authors declare no conflict of interest.

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